Clinical Supply Chain: Product Requirements Document (PRD)

Executive Summary

The Clinical Supply Chain (CSC) module delivers end-to-end clinical supply chain management on the Arc platform, covering reservation, dose ordering, batch planning, shipments, quality control, and chain-of-custody workflows. The module is built from the ARROW solution as a functional baseline and extended to support cell & gene therapies and other next-generation modalities.

Phase 1 has two parallel workstreams:

- **Demo-Ready NGT** — A rapid UI reskin and sanitized data swap enabling a credible client-presentable demo by XX/XX
- **Cell & Gene Prototype** — A Figma/video prototype capturing apheresis, processing, and return flows to support GTM conversations.

Success criteria: A reusable, white-label Arc module re-skinnable within days; full regulatory traceability (audit trail, e-signatures) carried from ARROW; a compelling cell & gene prototype for investor pitches; and a clear, costed roadmap for full data-model integration post-demo.

1. Introduction

1.1 Purpose

This PRD defines requirements for the Clinical Supply Chain Module to be delivered on the Arc platform. It establishes scope, functional requirements, user roles, constraints, risks, and delivery milestones for Phase 1.

1.2 Primary Goals

- Deliver a reusable, configurable, and white-label-ready clinical supply chain module on Arc.
- Produce a demo-ready NGT experience (UI reskin + sanitized data) by Week 4–5.
- Define and prototype cell & gene process differences (apheresis/processing/return flows) for GTM use.
- Ensure all production workflows meet regulatory requirements for audit trails and e-signatures.

1.3 Problem Statement

The existing ARROW solution is tightly coupled to a single client (RayzeBio). This PRD defines the work required to:

- **Productize** ARROW into a configurable Arc module, decoupling client-specific schema and branding.
- **Accelerate GTM** by enabling rapid demo delivery through UI reskinning and dummy data, without waiting for full backend refactoring.
- **Expand addressable market** by capturing cell & gene workflow requirements and delivering an early prototype for investors and client conversations.

2. Functional Requirements

2.1 Centralized Dashboard

#	Requirement	Priority
2.1.1	The system shall provide role-specific home pages for: Clinic Site, Clinical Operations, Planning, Manufacturing, Logistics, Quality, and Admin.	**MUST**
2.1.2	Each home page shall display contextual KPI counts (open SARFs, pending Dose Orders, at-risk patients) and quick-action entry points.	**MUST**
2.1.3	The system shall provide dosing, manufacturing/delivery, and shipment calendars filterable by study, product, and site.	**MUST**

2.1.4	The system shall display a persistent red at-risk banner whenever patients or processes are flagged at-risk.	**MUST**
2.1.5	*Demo scope:* Dashboards and calendars shall be populated with sanitized RayzeBio data within the reskinned Arc UI.	**MUST**

Commented [PC1]: Replace RayzeBio - Oak Avenue Pharma

2.2 Reporting & Analytics

#	Requirement	Priority
2.2.1	The system shall support embedded BI tiles (Sigma or equivalent) configurable per client and per role.	**MUST**
2.2.2	Analyst users shall have access to a SQL editor and the ability to export data as .xlsx for offline audit and review.	**MUST**
2.2.3	The system shall generate exportable audit trail reports including e-signature views sufficient for regulatory inspection.	**MUST**

2.3 Permissions & Access Control

#	Requirement	Priority
2.3.1	The system shall enforce role-based access control (RBAC) with row-level data filtering and configurable role archetypes.	**MUST**
2.3.2	Administrators shall be able to create, modify, and deactivate role archetypes and user permissions through an admin UI without engineering intervention.	**MUST**
2.3.3	The system shall support non-person accounts for AI services and system integrations.	**MUST**
2.3.4	The system shall provide admin reports showing current user/role assignments and enabling historical role reconstruction for audit purposes.	**MUST**

2.4 White-Label & Demo Mode

#	Requirement	Priority
2.4.1	The system shall support full UI white-labeling: configurable themes, logos, labels, and terminology (including replacement of all "RayzeBio" and "ARROW" references).	**MUST**
2.4.2	String replacements shall apply to both front-end display text and background system strings.	**MUST**
2.4.3	The system shall support a sanitized demo mode in which dummy/synthetic data is ingested and displayed, ensuring no proprietary client data is exposed in GTM contexts.	**MUST**
2.4.4	Demo mode shall be operable in non-production environments only, with a configuration flag preventing accidental activation in PROD.	**MUST**

2.5 Core Clinical Workflows

- **SARF:** Creation by Clinic Site; review/approval by Clinical Operations.
- **Dose Orders:** Submission by Clinic Site; approval and management by Clinical Operations.

- **Batch Planning:** Creation and management by Planning; integration with radioisotope planning and inventory sandbox.
- **MARK AI Allocation:** Human-in-the-loop dose allocation recommendations with full traceability; alerts on allocation failures.
- **Shipments:** Creation by Logistics; temperature log recording; shipping holiday visibility.
- **Quality Control:** CoA upload; e-signature approvals; quality status management.
- **Chain of Custody:** Receipt and administration confirmation by Clinic Site.
- **Dose Destruction:** Which is initiated in case of a bad batch being delivered to the clinic for administration.

3. Non-Functional Requirements

3.1 Performance & Scalability

- UI screens shall load within **3 seconds** under normal operating conditions.
- MARK AI allocation runtimes shall complete within defined SLA thresholds (to be established by Engineering in Week 0–1).
- The system shall be designed to scale to thousands of concurrent users without degradation.

3.2 Data Integrations

Integration	Method	Frequency
IRT / RTSM	API	Real-time
SAP	Batch	Nightly
ARMADA	Batch	Nightly
CTMS / EDC	Read-only views	Configurable
Annual plans & holidays	Bulk upload	As needed

3.3 Environment & Deployment

- The system shall maintain: **STAGING, TEST, UAT, PRE-PROD, PROD** environments.
- PROD shall have SQL replication enabled for disaster recovery.
- A documented, repeatable process shall exist for non-production environment to refresh from PROD.

3.4 Regulatory & Compliance

- The system shall maintain a full, tamper-evident audit trail for all data changes (user, timestamp, before/after values).
- All regulated actions shall support **21 CFR Part 11-compliant e-signatures** with metadata stored and displayable.
- All timestamps shall be time zone-aware and stored in UTC, with local time zone display.
- Regulatory controls shall be carried forward from the validated ARROW solution to minimize re-validation scope.

4. User Roles & Permissions

Role	Key Capabilities
Admin	Configure archetypes/permissions, user provisioning, non-person accounts
Analyst	Reporting, SQL editor, data exports
Manager	Quality and metadata dashboards
Client Viewer	Configurable BI tiles, read-only access

Clinic Site	Create SARFs, submit Dose Orders, confirm receipt/administration
Clinical Operations	Review/approve SARFs & Dose Orders, manage SARF workflow
Planning	Batch creation, radioisotope planning, inventory sandbox, MARK confirmation
Manufacturing	Manage machines/blackout dates, view batches
Logistics	Create shipments, record temperature logs, view shipping holidays
Quality	CoA upload, e-signature approvals, quality status management

Commented [SS2]: Consider including more context to view batches: Map planned manufacturing to actual demand from enrolled patients; View supply and planned manufacturing

5. Assumptions & Constraints

Assumptions

- Arc will host the module on a Mainsail-configured application consistent with the ARROW architecture.
- Phase 1 demo scope is limited to UI reskin and sanitized data swap; backend schema renames are post-demo.
- The cell & gene rapid prototype will be produced outside Arc (Figma, low-code, or video) to meet pitch timelines.
- Production releases must meet all ARROW regulatory controls for audit trails and e-signatures.

Constraints

- The ARROW backend contains client-tied column names and hardcoded strings that must be refactored before full productization — explicitly out of scope for Phase 1.
- Internal tooling and install permissions may limit rapid prototyping options; external vendor engagement may be required.
- The target demo date (Feb 24/25) is fixed and drives Phase 1 scope decisions.

6. Open Questions

#	Question	Owner	Due
OQ-1	What level of data sanitization is permitted for demos — UI strings only, or full schema rename?	Patrick / Srinivas	Week 1
OQ-2	Who owns sign-off on cell & gene workflow definitions?	Dennis	Week 1
OQ-3	Which tool and owner will produce the cell & gene prototype (KD / Pradeepta / vendor)?	Vishal / Jason	Week 1
OQ-4	What are the target manufacturers and products beyond NGT for the GTM list?	Vishal / Jason	Week 2
OQ-5	What are the exact security and compliance requirements for multi-client PROD?	Srinivas / QA	Week 2
OQ-6	Effort estimate (hours by role) for: (a) UI reskin + data swap, and (b) full data-model integration.	Srinivas	Week 1

7. Risks & Mitigation

Risk	Likelihood	Impact	Mitigation

Backend refactor scope larger than estimated	**MEDIUM**	**HIGH**	Phase delivery: demo = UI + data-swap only. Produce schema delta and role-hour estimates before committing to full integration.
Rapid prototyping blocked by tooling/permissions	**MEDIUM**	**MEDIUM**	Engage KD or external vendor to produce 1–2 min Figma/video prototype within 48–72 hrs of kick-off.
Regulatory/validation blocking PROD release	**LOW**	**HIGH**	Initiate QA/Validation review in Week 1; carry forward ARROW validated e-signature controls.
Misalignment on cell & gene requirements	**MEDIUM**	**MEDIUM**	Host workshop(s) with Dennis and Srinivas to produce process maps and prioritized backlog.
Demo data IP / confidentiality exposure	**LOW**	**MEDIUM**	Enforce sanitized demo mode; limit demo data to non-production environments only.

8. Milestones & Timeline

Phase 1 Kick-Off
Target: Week 0
Patrick / Vishal / Jason

Clinical Workflow Maps (RLT + CAR-T)
Target: Week 0–1
Dennis

Effort Estimate by Role
Target: Week 0–1
Srinivas

First Draft PRD
Target: Week 1
Patrick

Lo-Fi UI Draft + Cell & Gene Prototype
Target: Week 1–2
KD

Dev: UI Reskin + Data Swap
Target: Week 2–4
Srinivas

Solution Lead Approval
Target: Week 4
Vishal / Jason

UAT for Demo
Target: Week 4–5
QA + Clinical SMEs

🎯 **Demo to Bobby / Investor Pitch**
Target: Feb 24/25
Patrick / Vishal / Jason

Data-Model Refactor Planning
Target: Post-Demo
Engineering

8. Approval and Sign-Off

Business Owner: [Vishal / Jason / Meredith] — Sign-off required for go/no-go on product investment
Technology Owner: Srinivas - feasibility & effort sign-off; Dan Cardinal for executive sponsorship

Appendices

Appendix A: Cross-Functional Stakeholders

Name	Role	Responsibility
Patrick	Product	PRD owner & GTM liaison

Vishal	Functional SME & Owner	
Srinivas	Engineering	Effort estimates and delivery
Dennis	Technical SME	Provide experience with client workflows (Legend Biotech)
KD	UX / Design	Rapid prototype and lo-fi UI
Jason / Vishal	Technical SME	Target clients and pitch materials
Quality Team	QA / Validation	Compliance review and UAT

Appendix B: Deferred / Future Considerations

- Two-way validated integrations from Arc to external systems.
- Support for plasmid and gene-editing modalities.
- Advanced AI allocation enhancements for commercial scaling.
- Full backend schema decoupling (client-agnostic data model).

Appendix C: Reference Documents

- ARROW User Requirements Specification (URS): baseline for functional and regulatory requirements.
- ARROW Validation Documentation: carried forward for e-signature and audit trail compliance.
- ARROW Data Model: ARROW Data Model - 20250701 STTM - ANW NA (1).xlsx
- Cell and Gene Therapy Logistics Management: cell-and-gene-therapy-logistics-management.pdf
- Case study: Beghou RadioconjugateCase Study.pdf

Appendices

Appendix A: Cross-functional stakeholders

- Product (Patrick): PRD owner & GTM liaison.
- Clinical SME (Dennis): Domain owner for cell & gene workflows.
- Engineering (Srinivas): effort estimates and delivery.
- UX/Design (KD): rapid prototype.
- Revenue/GTM (Jason/Vishal): target clients & pitch materials.
- QA/Validation (Quality): compliance and UAT.

Appendix B — Requirements deferred / future considerations

- Two-way validated integrations from Arc to external systems (future).
- Support for plasmid/gene-editing modalities.
- Advanced allocation enhancements for commercial scaling.

Appendix C — Deep Dive Research

- Baseline requirements: ARROW User Requirements Specification (URS).